

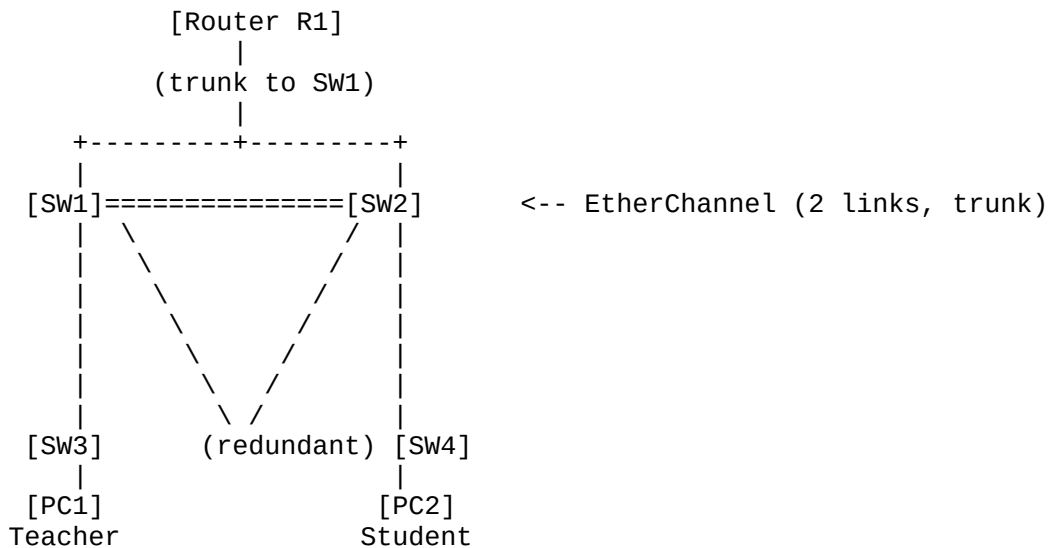
Packet Tracer Lab: VLANs, STP, and EtherChannel (No L3 Switches)

Topology Overview

Devices needed:

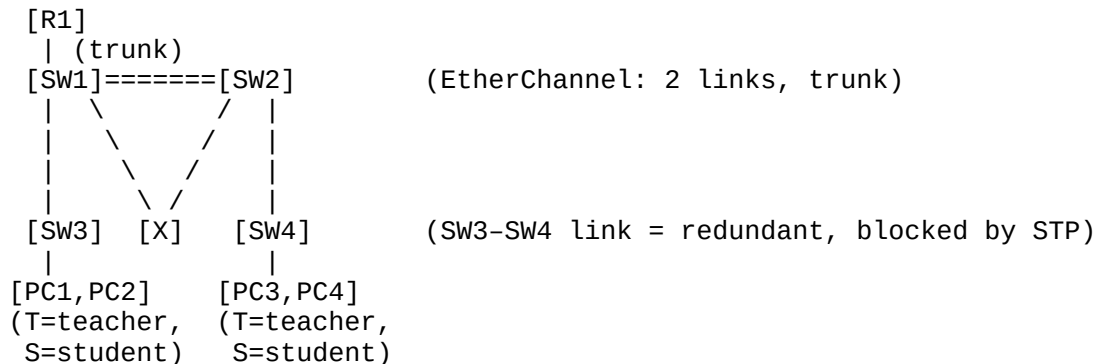
- 4 × Cisco 2960 switches (SW1–SW4)
- 1 × Router (2911 or 4331) for inter-VLAN routing (router-on-a-stick)
- 4 × PCs (2 per VLAN)

text



Simplified recommended layout:

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Cabling:

- R1 Gi0/0 → SW1 Fa0/24 (trunk)

- SW1 Fa0/1 ↔ SW2 Fa0/1 (EtherChannel member 1)
- SW1 Fa0/2 ↔ SW2 Fa0/2 (EtherChannel member 2)
- SW1 Fa0/3 → SW3 Fa0/24 (trunk)
- SW2 Fa0/3 → SW4 Fa0/24 (trunk)
- SW3 Fa0/23 ↔ SW4 Fa0/23 (redundant trunk – STP will block one end)
- PC1 (Teacher) → SW3 Fa0/1
- PC2 (Student) → SW3 Fa0/2
- PC3 (Teacher) → SW4 Fa0/1
- PC4 (Student) → SW4 Fa0/2

VLAN Plan

VLAN	Name	Subnet	Gateway
10	TEACHERS	192.168.10.0/24	192.168.10.1
20	STUDENTS	192.168.20.0/24	192.168.20.1

STP: Use Rapid-PVST. Make **SW1 the root for VLAN 10** and **SW2 the root for VLAN 20** (load balancing).

Router R1 Configuration (Router-on-a-Stick)

text

```
enable
configure terminal
hostname R1
```

```
interface g0/0
no shutdown
exit
```

```
interface g0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
exit
```

```
interface g0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
exit
```

```
! Optional: enable routing (on by default)
end
write memory
```

SW1 Configuration (Root for VLAN 10)

text

```
enable
configure terminal
hostname SW1
```

```
! Create VLANs
vlan 10
  name TEACHERS
vlan 20
  name STUDENTS
exit
```

```
! VTP transparent (so no VTP domain issues in PT)
vtp mode transparent
```

```
! Access ports (none on SW1 in this design, but include for safety)
! Trunk to Router
interface fa0/24
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit
```

```
! Trunk to SW3
interface fa0/3
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
exit
```

```
! EtherChannel to SW2
interface range fa0/1 - 2
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  channel-group 1 mode active
  no shutdown
exit
```

```
interface port-channel 1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
exit
```

```
! STP: Rapid-PVST, root for VLAN 10
spanning-tree mode rapid-pvst
spanning-tree vlan 10 root primary
spanning-tree vlan 20 root secondary
```

```
end
write memory
```

SW2 Configuration (Root for VLAN 20)

text

```
enable
configure terminal
hostname SW2

vlan 10
  name TEACHERS
vlan 20
  name STUDENTS
exit
vtp mode transparent

! Trunk to SW4
interface fa0/3
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  exit

! EtherChannel to SW1
interface range fa0/1 - 2
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  channel-group 1 mode active
  no shutdown
  exit

interface port-channel 1
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  exit

! STP: Rapid-PVST, root for VLAN 20
spanning-tree mode rapid-pvst
spanning-tree vlan 20 root primary
spanning-tree vlan 10 root secondary

end
write memory
```

SW3 Configuration (Access switch – Teacher & Student PCs)

text

```
enable
configure terminal
hostname SW3

vlan 10
  name TEACHERS
vlan 20
  name STUDENTS
exit
```

```
vtp mode transparent

! PC1 - Teacher
interface fa0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
  exit

! PC2 - Student
interface fa0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
  exit

! Trunk to SW1
interface fa0/24
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  exit

! Redundant trunk to SW4 (STP will block one side)
interface fa0/23
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  exit

spanning-tree mode rapid-pvst
end
write memory
```

SW4 Configuration (Access switch – Teacher & Student PCs)

```
text

enable
configure terminal
hostname SW4

vlan 10
  name TEACHERS
vlan 20
  name STUDENTS
exit
vtp mode transparent

! PC3 - Teacher
interface fa0/1
  switchport mode access
  switchport access vlan 10
  no shutdown
  exit
```

```

! PC4 - Student
interface fa0/2
  switchport mode access
  switchport access vlan 20
  no shutdown
  exit

! Trunk to SW2
interface fa0/3
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  exit

! Redundant trunk to SW3
interface fa0/23
  switchport mode trunk
  switchport trunk allowed vlan 10,20
  no shutdown
  exit

spanning-tree mode rapid-pvst
end
write memory

```

PC IP Configuration

PC	VLAN	IP	Mask	Gateway
PC1	10	192.168.10.10	255.255.255.0	192.168.10.1
PC2	20	192.168.20.10	255.255.255.0	192.168.20.1
PC3	10	192.168.10.11	255.255.255.0	192.168.10.1
PC4	20	192.168.20.11	255.255.255.0	192.168.20.1

In Packet Tracer: **Desktop** → **IP Configuration** → **Static**.

Verification Commands

EtherChannel:

text

```

show etherchannel summary
! Port-channel 1 (SU) with Fa0/1 and Fa0/2 (P)

```

STP load balancing:

text

```

show spanning-tree vlan 10
! SW1 should show "This bridge is the root"
show spanning-tree vlan 20

```

```
! SW2 should show "This bridge is the root"
show spanning-tree
! On SW3/SW4, one uplink should be "Blocking/Alternate" for each VLAN
```

VLAN / Trunk:

text

```
show vlan brief
show interfaces trunk
```

Connectivity:

- PC1 → PC3 (same VLAN, different switches): **should ping**
 - PC1 → PC2 (different VLANs, via router): **should ping**
 - Unplug one EtherChannel link: traffic continues (redundancy test)
 - Unplug SW1–SW3 trunk: STP converges through SW4
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What Students Should Observe

1. **EtherChannel** bundles two physical links into one logical link — `show etherchannel summary` shows SU (in use) and P (bundled).
 2. **STP** blocks the SW3–SW4 redundant link (one end in Alt/Blocking state) — check `show spanning-tree`.
 3. **Per-VLAN load balancing** — SW1 root for VLAN 10, SW2 root for VLAN 20.
 4. **Router-on-a-stick** enables inter-VLAN communication without an L3 switch.
 5. **Fault tolerance** — pull any single link and connectivity survives.
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Common Student Pitfalls

- Forgetting `switchport trunk allowed vlan 10,20` → VLANs pruned.
- Mismatched EtherChannel modes (use **active/active** = LACP or **desirable/desirable** = PAgP; never on/on with PT bugs).
- Forgetting `no shutdown` on port-channel members.
- Wrong STP priority if using `root primary/secondary` — verify with `show spanning-tree`.
- Native VLAN mismatch → CDP warnings; keep native VLAN 1 on both ends.